■■ CTPPO

Cyber Threat Prioritization & Path Optimization

AI-Powered CVE Severity Classification & Attack Path Analysis

THE PROBLEM

Security teams are drowning in vulnerability data. With **25,000+ new CVEs published annually** and growing, manual prioritization is impossible. Critical vulnerabilities slip through while teams waste time on low-priority issues.

25,000+	73%	$4.45M
New CVEs per year	Security teams understaffed	Avg. cost of data breach

OUR SOLUTION

CTPPO uses **multi-modal deep learning** to automatically classify CVE severity and identify optimal attack paths. Our system combines natural language understanding with structured vulnerability data to deliver accurate, explainable predictions.

- ■ **Multi-Modal Classification:** Combines text descriptions with CVSS components, CWE types, and exploit indicators
- ■ **Explainable AI:** Attention visualization shows WHY each prediction was made
- ■■ **Attack Path Analysis:** NAMOA* algorithm finds ALL Pareto-optimal attack paths
- ■ **Real-Time Processing:** Analyze new CVEs as they're published

KEY METRICS

70.5%	189K	8	93%

F1 Score	CVEs Processed	CVSS Features	Coverage

TECHNOLOGY STACK

Component	Technology	Purpose
Text Analysis	DistilBERT	NLP understanding of CVE descriptions
Multi-Modal Fusion	PyTorch	Combines text + structured features
CVSS Processing	Custom Encoders	8 vulnerability component features
Attack Paths	NAMOA*	Multi-objective path optimization
Explainability	Attention Viz	Transparent decision-making

COMPETITIVE ADVANTAGE

Feature	Traditional Tools	CTPPO
Input Data	Text only	**Multi-modal (text + 8 CVSS + CWE)**
Explainability	Black box	**Full attention visualization**
Attack Paths	Single path or none	**ALL Pareto-optimal paths**
Label Quality	Inconsistent NVD	**CVSS-score derived**
Customization	Fixed rules	**Trainable on your data**

USE CASES

- **Security Operations Centers (SOC):** Automate CVE triage and prioritization
- **Vulnerability Management:** Focus remediation on highest-risk vulnerabilities
- **Penetration Testing:** Identify optimal attack paths for red team exercises
- **Compliance:** Document risk-based prioritization decisions

ROADMAP

Phase	Timeline	Deliverables
Current	Jan 2026	70.5% F1 model, NAMOA* integration
Next	Feb 2026	78-82% F1 with CVSS components
Beta	Q2 2026	REST API, web dashboard
Production	Q3 2026	Enterprise deployment, integrations

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"Transforming vulnerability management with AI-powered intelligence"